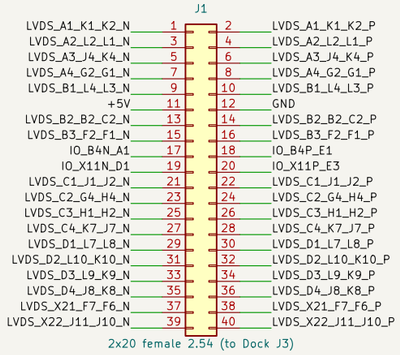
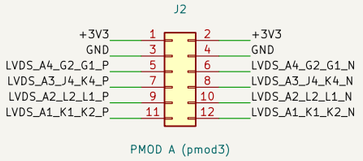
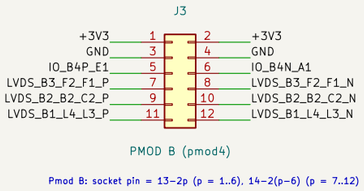
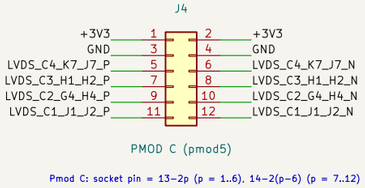
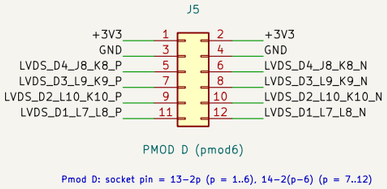
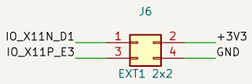


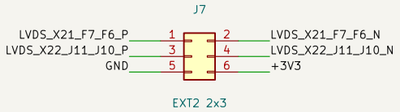


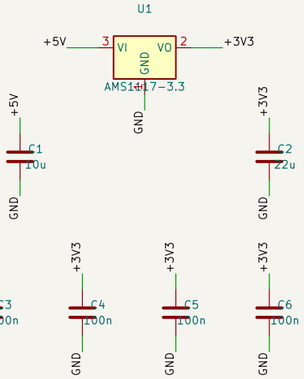


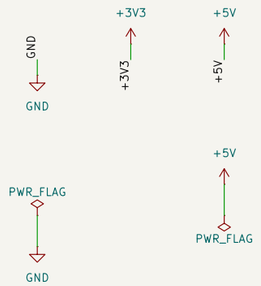












Tang Primer 25K Dock 40-pin header (J3 on the Dock, DE0 compatible) -> 4 Pmod host sockets (pmod3..pmod6).

Header pins 11 = +5V, 12 = GND; no 3.3 V pin: U1 makes the Pmods 3.3 V from +5V.

Pins (2k-1, 2k) are GW5A true LVDS pairs (even = P, odd = N), except 17/18 (A1/E1) and 19/20 (D1/E3).

Each pair lands on a Pmod column: P = Pmod pin n, N = pin n+6. LVDS_* nets are routed as 100 ohm pairs.

Pmod B column 4 carries A1 / E1 (single-ended); EXT1 = D1, E3; EXT2 = F7/F6 and J11/J10 pairs.

Bank 2 pairs (A, B1, C1) have no on-chip differential termination on GW5A (use external 100 ohm for LVDS inputs).

Kenta Ida (@ciniml)

fpga_samples / eda/targets/tangprimer25k

Sheet: /

File: DE0Header_PmodAdapter.kicad_sch

Title: Tang Primer 25K 40-pin header to Pmod adapter

Size: A3	Date: 2026-10-20	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1